

INNOVATIVE PRODUCT DEVELOPMENT

FINAL REPORT

GROUP 01

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- **Introduction**

- A height and angle adjustment desk is a workstation designed to provide ergonomic benefits by allowing users to modify the height and tilt of the desk surface. These desks help improve posture, reduce strain, and enhance productivity by accommodating various working positions, including sitting and standing.



- To be effective, a height and angle adjustment desk must include key features such as a reliable height adjustment mechanism, which can be manual (lever/crank) or electric, and an angle adjustment feature for improved usability. A stable and durable building, often using metal or strong wood, ensures longevity and safety. It

should also have a sufficient weight capacity to support devices like monitors, laptops, and other office equipment. Lastly, user-friendly designs with smooth transitions between settings make these desks more practical for everyday use.

- Several commercial products are available to meet different needs and budgets. The Uplift V2 Standing Desk offers stability and extensive customization options, while the varidesk Pro Plus 36 serves as a desktop converter for sit-stand flexibility. The is Flexispot EC1 an affordable electric standing desk, whereas the Seville Classics Airlift Pro features motorized height adjustments with an ergonomic design. For those seeking smart functionality, the Autonomous Smart Desk provides AI-powered adjustability with pre-set positions. These desks cater to various workspaces, ensuring a healthier and more adaptable workstation setup.



Autonomous smart desk



Flexispot EC1

• Opportunity Identification

- The scope of the project focuses on developing an innovative, engineered product that automates and optimizes repetitive tasks, reducing manual workload while enhancing efficiency and productivity. The product will be designed to address tasks that require physical intervention, monitoring, or repetitive actions, making it suitable for diverse work environments. A key requirement is the incorporation of at least one moving part while ensuring cost-effective manufacturing using available faculty resources.
- The primary goal is to create a cost-effective, user-friendly automation solution that simplifies time-consuming tasks. The success of the product will be measured by its ability to reduce task completion time by at least 30%, receive positive user feedback for usability and effectiveness, and maintain cost-efficient production. The project ultimately aims to develop a practical and innovative solution that improves user experience and operational efficiency.

▪ 10 promising ideas

1. Height and Angle Adjustable Desk.
2. Adjustable Phone Stand.
3. Retractable Garden Hose Wheel.
4. Collapsible Ladder.
5. Motorized Camera Slider.
6. Foldable Treadmill.
7. Rotating Spice Rack.
8. Adjustable Desk Lamp.
9. Portable BBQ Grill.
10. Popup Power Outlet.

▪ Screen Opportunities

PROJECT MANAGEMENT SCREEN OPPORTUNITIES					
	Market Demand	Technical Feasibility	Innovation & Uniqueness	Cost Effectiveness	Long Term Profitability
HEIGHT & ANGLE ADJUSTABLE DESK	High (Home, Offices, School)	Can be manufactured in Faculty.	Dual adjustment for ergonomics	Mid Range Cost, High Perceived value	Strong Demand, Long- Term growth
ADJUSTABLE PHONE STAND	Moderate (Home, Offices, School)	Simple hinge/ joint mechanism.	Common Design, Little differentiation.	Low production cost	Market is saturated
RETRACTABLE GARDEN HOSE REEL	Popular among gardeners	Spring/motorized system	Many similar products exist	Moderate, but competitive pricing	Seasonal demand, limited scalability
COLLAPSIBLE LADDER	Essential for home/construction use	Telescoping sections feasible	High competition, minor changes possible	High material cost	Competition limits profit
MOTORIZED CAMERA SLIDER	High (photographers & videographers)	Complex motorized precision needed	Unique for smooth filming	High precision & motor costs	Growing interest in videography tools
FOLDABLE TREADMILL	Growing fitness trend	Requires high-quality running surface & motor	Foldability is a key space-saving feature	Expensive materials & motors	Expanding home fitness market

	Market Demand	Technical Feasibility	Innovation & Uniqueness	Cost Effectiveness	Long Term Profitability
ROTATING SPICE RACK	Low (many alternatives)	Simple rotating base	Simple & widely available	Low-cost plastic or wood	Low margins, many alternatives
ADJUSTABLE DESK LAMP	Common for workspaces	Easy hinge-based design	Standard adjustable lamps exist	Simple & affordable	Highly competitive, low profit
POPOP POWER OUTLET	Growing demand in modern offices	Spring-loaded/electric system	Blends into furniture when not in use	Mid-range cost, high demand	Integration with smart homes increasing
PORTABLE BBQ GRILL	Seasonal, but popular	Foldable metal/heat-resistant parts	Compact grilling solution	Good balance of cost & usability	Strong demand, stable market

Select Exceptional Opportunities:

Criteria	Height and Angle Adjustable desk	Collapsible Ladder	Hose Reel
Is it real?	✓ Strong demand in office, home and education sectors	✓ Needed in home and construction work	✓ Useful for homeowners and gardeners
Market Size and Demand	✓ Large and growing due to WFH trends	✓ High demand but niche	✓ High demand but seasonal Technical Feasibility
Technical Feasibility	✓ Can be manufactured with faculty resources (metal frames, motors, hinges)	✓ Uses telescopic locking mechanisms	✓ Requires spring/motorized retraction system
Can We Win?	✓ Unique due to dual height & angle adjustment	✗ Many existing collapsible ladders	✗ Many retractable hose reels available Competitive Advantage
Competitive Advantages	✓ Ergonomics, multi-use for sitting/standing/drafting	✗ Market already has similar products	✗ Not significantly different from current solutions
Is it Worth it?	✓ Strong long-term potential in home offices & workspaces	✗ Competition is high, limited differentiation	✗ Seasonal sales, limited innovation potential
Cost and Profitability	✓ Mid-range cost, high perceived value	✗ High material cost, limited price flexibility	✗ Moderate cost but competitive pricing pressures

- In contrast, while the Collapsible Ladder and Retractable Garden Hose Reel have their own merits, they face higher competition, less differentiation, and more challenges in terms of cost and profitability.
- Therefore, focusing on the Height and Angle Adjustable Desk for modeling appears to be the most promising and strategic choice.

- **Mission Statement**

- **Product Overview:** This adjustable desk is designed to enhance comfort and productivity by allowing users to modify both the height and angle of the work surface. It provides flexibility for various tasks, such as working while sitting or standing, reading, writing, and creative activities.
- **business objectives:** focus on introducing the product in the next development phase, improving task efficiency by reducing completion time by at least 30%, and establishing a strong presence in home offices and educational environments
- **target market:** primarily includes remote professionals and office workers, while secondary markets include students, educators, designers, and artists who require adaptable workspaces.
- **key considerations:** such as manual crank or electric motor adjustments, durable materials for stability and longevity, and cost-effective manufacturing using available resources.
- **relevant stakeholders:** include end-users like professionals, students, and creative workers, as well as retailers, distributors, sales and marketing teams, and manufacturing and production teams responsible for ensuring quality and efficiency. This project aims to create a functional, adaptable, and widely accessible ergonomic desk solution.



• Customer Need Analysis

Detailed Analysis Table

Question	Customer Statement	Need Statement	Features/Comments
Desk Height Range Preference	Most respondents (mostly 18–30 years, students/office workers) chose 66–76 cm as ideal for adults, and a few indicated ranges like 50–55 cm or 55–63 cm for children.	The desk should be adaptable to a number of different user groups, by providing a variety of height options for adults and children.	A modular design with adjustable height options, or designs catered for specific age groups.
Adjustable Surface Angle	A majority of responders indicated a desire for an adjustable surface angle with a preference for medium tilt (15°–30°) and some of them for a small tilt (0°–15°); very few asked for a large tilt.	The desk should include an adjustable surface angle mechanism, at least it should have medium and small tilt options.	Integrated adjustable tilt mechanism with preset configurations to simplify user operation.
Desk Size	Most of the responders favored medium (120×75 cm) and large (150×80 cm) desk sizes, with only one instance of a custom size request.	With a particular focus on medium and large models to serve a vast workspace, the design ought to be comprised of several dimensions.	Offer standard sizes with an option for minor customizations if required.
Material Preference	Responders mentioned materials such as MDF, wood, glass, metal, and polymer resin composite. Preferences varied but focused on durability and appearance.	The desk should be available in durable, high-quality materials that balance aesthetics with cost.	Options to choose between materials (e.g., MDF for affordability, wood for premium feel) along with finish customization.

➤ We visualized customer needs for clear and effective decision-making using these charts.

Height Adjustment Mechanism	Users were split between electric (motorized) and manual (crank system) adjustments.	Provide a choice between electric and manual height adjustment mechanisms to cater to varying preferences and budgets.	Consider a dual-option design or clearly differentiated product variants based on adjustment type.
Additional Features	Many responses included desired add-ons as built-in drawers/shelves, wheels for portability, USB charging ports, and wireless charging pads.	The desk should integrate additional features that enhance usability and convenience during work.	Incorporate built-in storage, mobility features (e.g., wheels), and integrated power/charging ports into the design.
Price Sensitivity	Most users were willing to spend between 10–25,000 LKR, with some indicating a lower budget (below 10,000 LKR) and a few of them suggesting higher ranges (25–50,000 LKR).	The product must be competitively priced to align with the majority's budget expectations while offering value through its features.	Devise a pricing strategy that aligns cost and features of both basic and advanced models.
Additional Comments	Some responders requested more color options and noted the importance of a lightweight yet robust design.	The desk needs to be offered in multiple colors and it needs to be made out of lightweight yet strong and durable materials at the same time.	In order for customers to express their individual aesthetic ideals in corresponding colors, manufacturers can provide essential lightweight materials in majority of products.

➤ Then we categorize needs using an affinity diagram to identify key themes such as cost, safety, demand and applications for our product.

❖ **Affinity Diagram**

Cost and Maintenance	Applications
Price should be reasonable and affordable.	Suitable for students needing multifunctional, ergonomic tables for studying and working.
Maintenance should be minimized	Ideal height adjustment range of 66–76 cm for adults
Durability and quality of materials are important to ensure longer lifespan and reduced maintenance costs.	Manual and electric height adjustment systems are equally in demand.
	Tables should offer flexibility with multiple material options:

Safety and Usability	Power Source
- Features that enhance usability - Drawers - Wheels for portability - USB ports - Wireless charging	Incorporate USB ports and wireless charging for modern tech integration.
Manual crank systems provide reliable adjustment without power dependency	Consider power-saving mechanisms for motorized tables.
Electric (motorized) tables offer ease of adjustment for quick transitions.	

Demand
Strong demand for tables that balance functionality with affordability
Students prioritize tables with versatile materials and essential features (drawers, USB ports, wireless charging).
High preference for tables with wheels to enhance mobility.

❖ Analytical Hierarchy process (AHP)

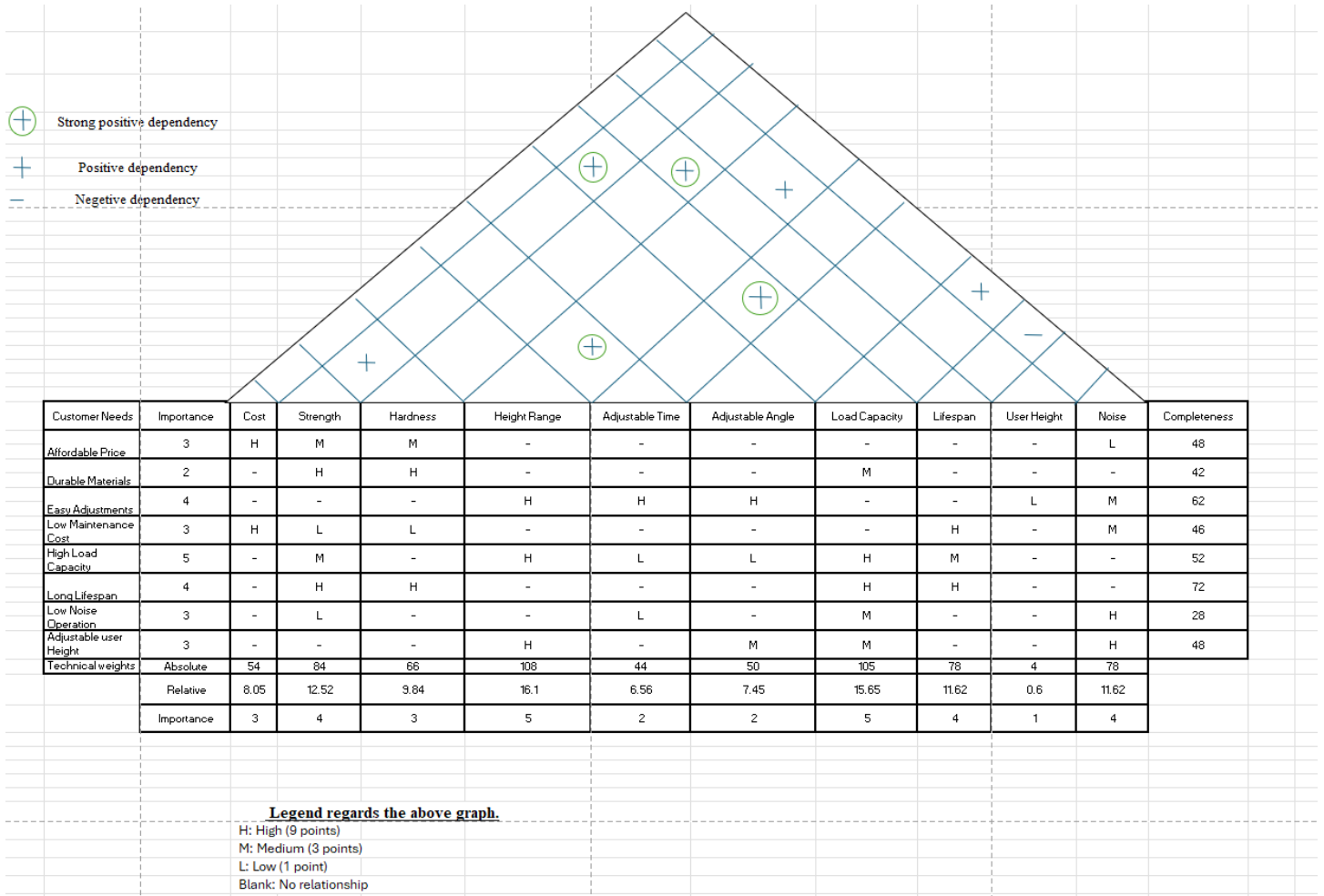
	cost and maintenance	Applications	Safety and Usability	Power Source	Demand	Row Total	Relative Importance	Scaled Importance
Cost and maintenance	1	0.2	0.2	5	1	7.4	10.54%	1.421
Applications	5	1	5	10	10	31	44.15%	5.000
Safety and Usability	5	0.2	1	10	10	26.2	37.32%	2.488
Power Source	0.2	0.1	0.1	1	1	2.4	3.40%	1.000
Demand	1	0.1	0.1	1	1	3.2	4.55%	1.182

• Rankings Of Needs

Need	Expected	Normal	Exiter	Status	Importance	Critical needs
Cost and maintenance		✗			1.421	✗
Applications			✗	New	5.000	✗
Safety and Usability			✗	Not difficult	2.488	✗
Power Source	✗			New	1.000	
Demand		✗			1.182	

• Product Specification

➤ QFD METHOD



➤ Product specification of Height and Angle Adjustable Desk.

Metric No.	Need Nos.	Matric	Importance	Unit	Approximate value
1.	1	Target Price	3	LKR	45 000/=
2.	1	Cost of material	3	LKR	25 000/=
3.	2	Material Tensile Strength	2	MPa	≥300
4.	2	Hardness	2	HRC	≥50
5.	3	Annual Maintenance Cost	3	LKR	2250/=
6.	4, 8	Hight Adjustment Range	4	cm	60-120
7.	4	Adjustment Time with One hand operation	4	S	≤5
8.	5	Angle Adjustability	4	Degree	0° to 60°
9.	5	Angle Locking Torque	4	Nm	≥ 50
10.	6	Load Capacity	5	kg	≥ 50
11.	7	Folded Thickness	3	cm	≤ 10
12.	7	Folded Volume	3	m ³	≤ 0.05
13.	9	Coefficient Of Friction	4		≥ 0.5
14.	10	Lifespan	4	Years	≥ 5
15.	11	Adjustment Force	4	N	≤ 5
16.	12	Total Weight	3	Kg	≤ 15
17.	13	Electric Motor Power	3	W	≤ 50
18.	13	Noise Level	3	dB	≤ 45
19.	14	Sizes to Carter to User Height	3	cm	140 to 200

➤ Competitive Benchmarking

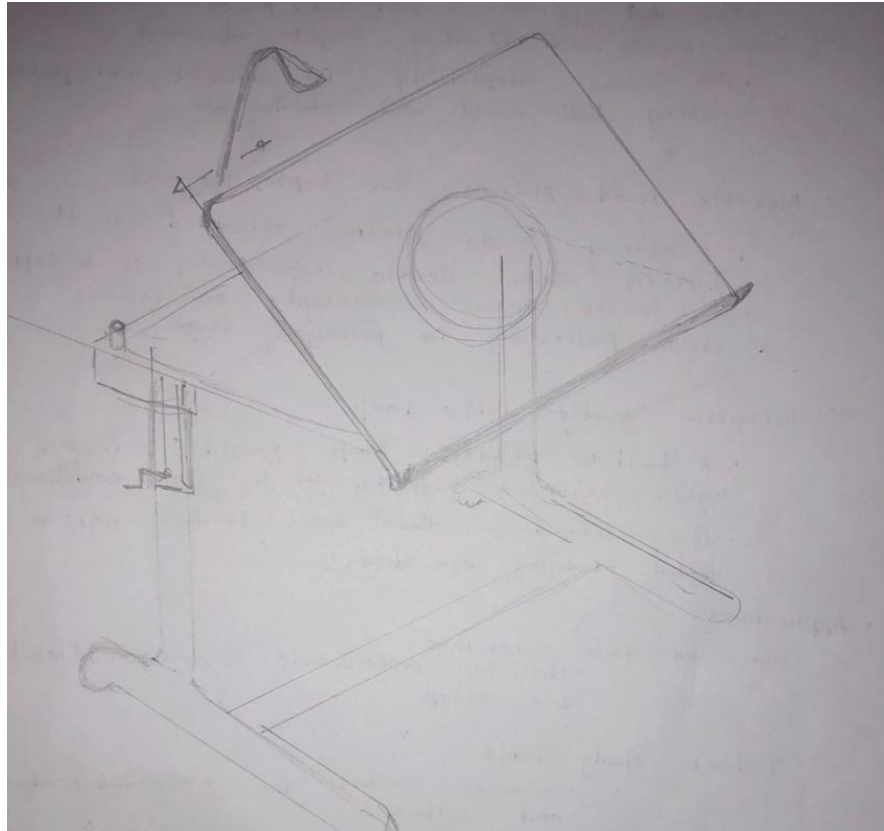
Metric No.	Need Nos.	Matric	Importance	Unit	Approximate value	Flexispot E2 pro standing desk	4 Ikea standing desk	Rocelco 32" standing desk	Tresanti adjustable standing desk
1.	1	Target Price	3	LKR	45,000	75, 000	65,000	40, 000	85 ,000
2.	1	Cost of material	3	LKR	25, 000	-	-	-	-
3.	2	Material Tensile Strength	2	MPa	≥300	350	310	280	320
4.	2	Hardness	2	HRC	≥50	55	52	48	53
5.	3	Annual Maintenance Cost	3	LKR	2,250	2,500	2,300	2,000	2,400

6.	4, 8	Hight Adjustment Range	4	cm	60-120	60-125	65-120	40-75	70-120
7.	4	Adjustment Time with One hand operation	4	S	≤5	≤4	≤5	≤6	≤4.5
8.	5	Angle Adjustability	4	Degree	0° to 60°	0°-60°	0°-50°	0°-45	0°-60°
9.	5	Angle Locking Torque	4	Nm	≥ 50	≥55	≥50	≥45	≥52
10.	6	Load Capacity	5	kg	≥ 50	100	70	50	55
11.	7	Folded Thickness	3	cm	≤ 10	≤8	≤9	≤10	≤9
12.	7	Folded Volume	3	m³	≤ 0.05	≤0.04	≤0.045	≤0.05	≤0.045
13.	9	Coefficient Of Friction	4	-	≥ 0.5	≥0.55	≥0.5	≥0.45	≥0.52
14.	10	Lifespan	4	Years	≥ 5	≥7	≥5	4	≥6
15.	11	Adjustment Force	4	N	≤ 5	≤4	≤5	≤5	≤4.5
16.	12	Total Weight	3	Kg	≤ 15	14	15	13	14.5
17.	13	Electric Motor Power	3	W	≤ 50	≤45	≤48	Non-motorized	≤46
18.	13	Noise Level	3	dB	≤ 45	≤45	≤50	N/A	≤44
19.	14	Sizes to Carter to User Height	3	cm	140 to 200	140-200	140-195	140-180	140-200



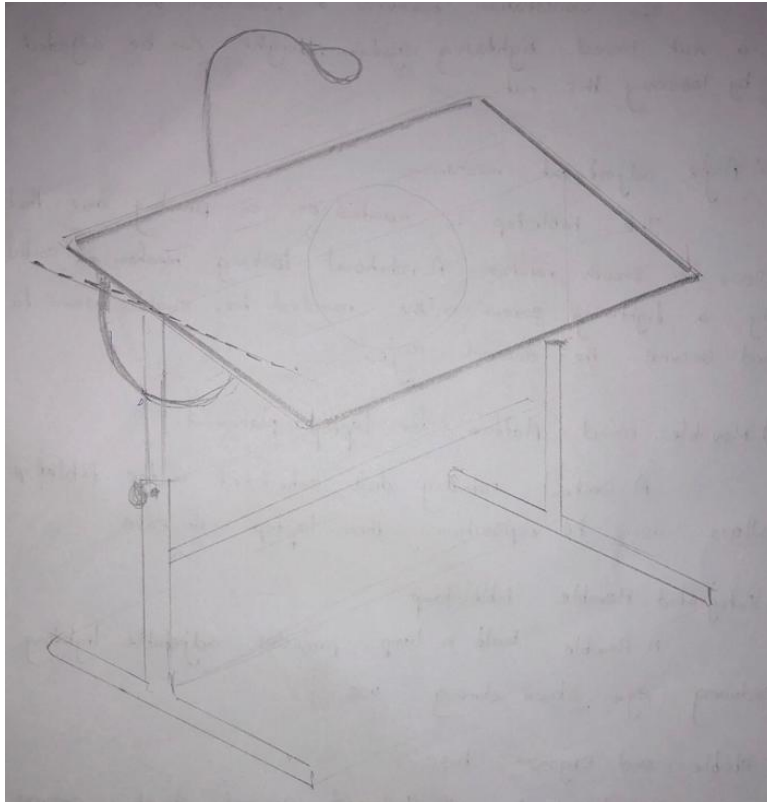
- **Concept Generation**

- ❖ *Concept A*



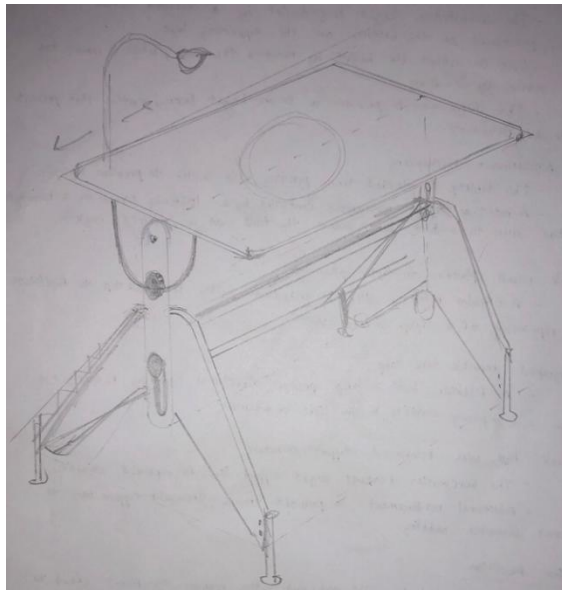
- The proposed workstation is a height- and angle-adjustable desk designed for ergonomic support and versatility. It features a rack and pinion mechanism for smooth height adjustment and a hinge-based system for modifying the tabletop angle. Additional functionalities include a rotating circular platform for flexible laptop positioning, an integrated movable table lamp for focused lighting, and a stable, ergonomic base for enhanced stability. This workstation is ideal for home and office use, students, and individuals with mobility challenges, providing comfort and adaptability for various work environments.

❖ Concept B



- Concept B presents a height- and angle-adjustable workstation designed for ergonomic flexibility and stability. It features a manual screw-based height adjustment using a nut-tightening system and a rotational locking mechanism for smooth angle modification. Additional features include a central rotating disk for flexible laptop placement, an integrated movable lamp for adjustable lighting, and a wide, stable base for secure usage. This workstation is ideal for home offices, students, and individuals with mobility needs, offering a practical and customizable workspace solution.

❖ Concept C



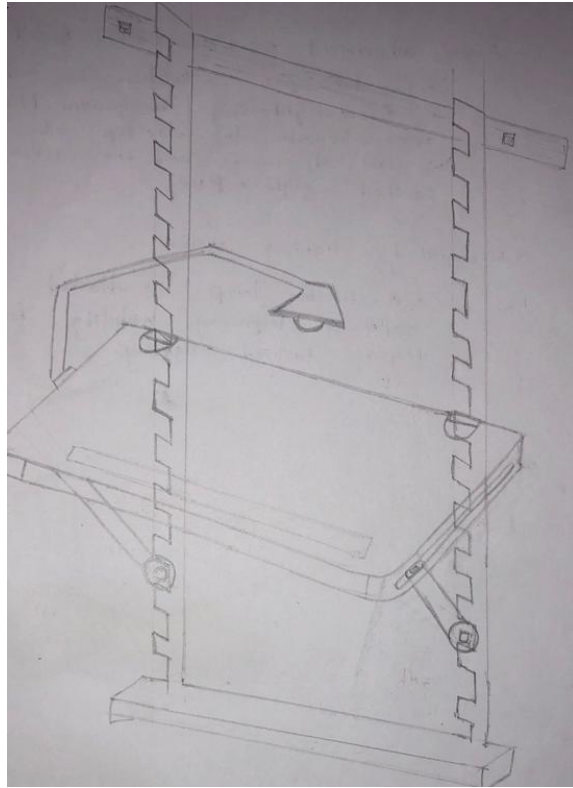
- Concept C introduces a height- and angle-adjustable workstation with a screw-nut height adjustment mechanism and a rotational locking system for angle control. It includes features like a movable round platform for laptop placement, an integrated adjustable lamp, and a stable reinforced base with angled support legs. However, this concept was rejected due to inefficiencies in the height adjustment mechanism, which required significant effort under load, was time-consuming, and prone to wear over time. Future designs should explore more user-friendly height adjustment systems like gas springs, hydraulic lifts, or ratchet-based locks for improved functionality.

❖ Concept D



- Concept D features a rack-and-pinion height adjustment system with a hand lever mechanism for smooth height changes and a spring-loaded support system for assisted angle adjustment. It includes an ergonomic frame design, an integrated flexible lamp, and a curved base for stability. However, this concept was rejected due to manufacturing complexity, potential stability issues, and durability concerns. The intricate mechanisms required precise machining, and long-term wear could affect functionality. Future designs should focus on simpler, more robust solutions like hydraulic or gas lift systems and ratchet-based angle locking mechanisms for better usability and cost-effectiveness.

❖ Concept E



- Concept E introduces a clip-based height adjustment system with preset notches for positioning and a screw-based angle adjustment mechanism for precise inclination control. It also features an integrated flexible lamp for better visibility. However, the concept was rejected due to structural weaknesses, limited height flexibility, and usability concerns. The clips could wear out or fail under heavy loads, and the preset height levels restricted smooth adjustments. Additionally, the manual screw-based angle adjustment was time-consuming and could loosen over time. Future designs should consider more durable alternatives like gas spring mechanisms or mechanical locking systems for improved reliability and ease of use.

❖ Concept F



- Concept F features a bevel gear-based height adjustment system with a hand lever for smooth and precise height changes, along with a toothed bar mechanism for secure angle adjustments. Additional features include a flexible lamp and a sturdy base with castor wheels for mobility. This design offers effortless height control, reliable angle locking, and enhanced durability, making it more user-friendly and ergonomic. However, it was noted for higher manufacturing complexity, increased weight, and potential maintenance needs. Future refinements should focus on optimizing production efficiency and reducing weight while maintaining the mechanical advantages of the system.

• Concept Selection

❖ Selection (Screening) Matrix using Pugh's Concept

- The table below presents the selection matrix comparing various product concepts against the reference concept. The evaluation is based on predefined selection criteria. Concept F has been selected as the most promising design.

Selection Criteria	Concept A	Concept B	Concept C (Reference)	Concept D	Concept E	Concept F
Ease of Handling	0	+	0	-	+	+
Interchangeability	0	0	0	0	0	+
Portability	0	0	0	0	+	+
Ease of Maintenance	+	0	0	0	0	+
Ease of Manufacture	-	+	0	0	-	+
Safety	0	-	0	-	+	+
Durability	0	+	0	0	+	+
Sum '+'	1	3	0	0	4	7
Sum '0'	5	3	7	6	2	1
Sum '-'	1	1	0	1	1	0
Net Score	0	2	0	-2	3	6

❖ Combine and Improve the Concepts

- The best aspects of Concept F have been improved based on the selection criteria to enhance functionality and usability.

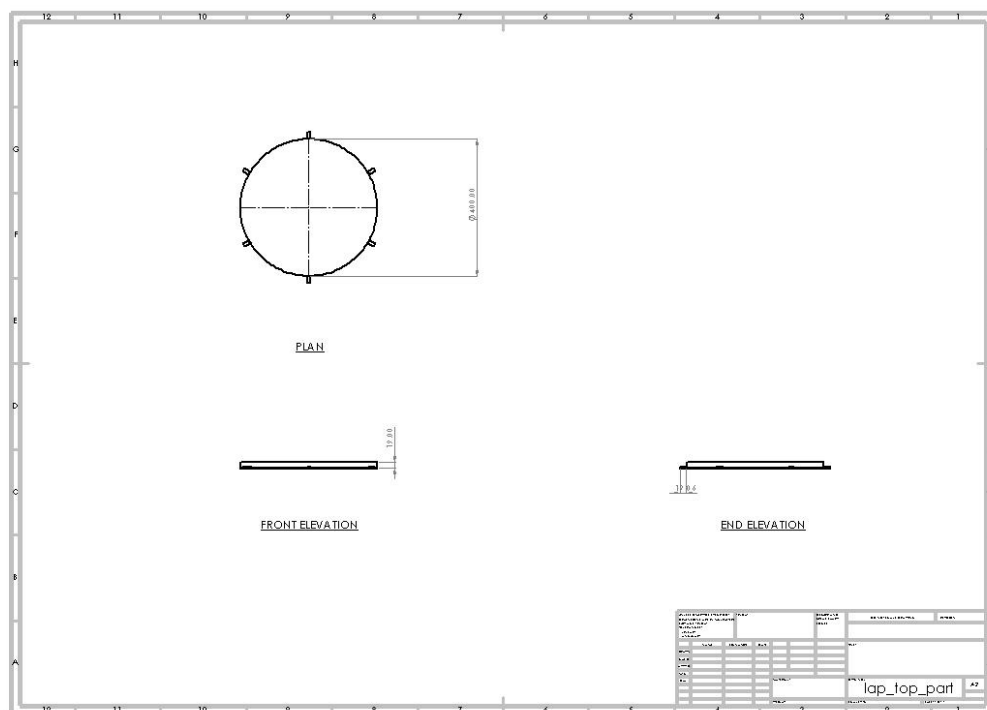
Selection Criteria	Weight	Rate (Concept A)	Weighted Score	Rate (Concept C)	Weighted Score	Rate (Concept F)	Weighted Score
Ease of Handling	20	2	0.4	3	0.6	5	1
Interchangeability	15	4	0.6	4	0.6	3	0.45
Portability	15	3	0.45	3	0.45	4	0.6
Ease of Maintenance	5	4	0.2	3	0.15	3	0.15
Ease of Manufacture	10	3	0.3	3	0.3	4	0.4
Safety	20	2	0.4	1	0.2	5	1
Durability	15	3	0.45	2	0.3	4	0.6
Total Score			2.8		2.6		4.2

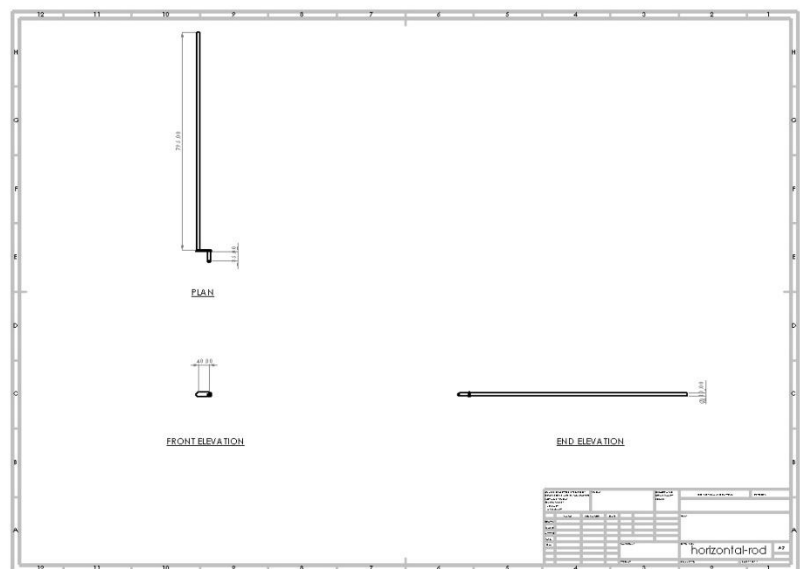
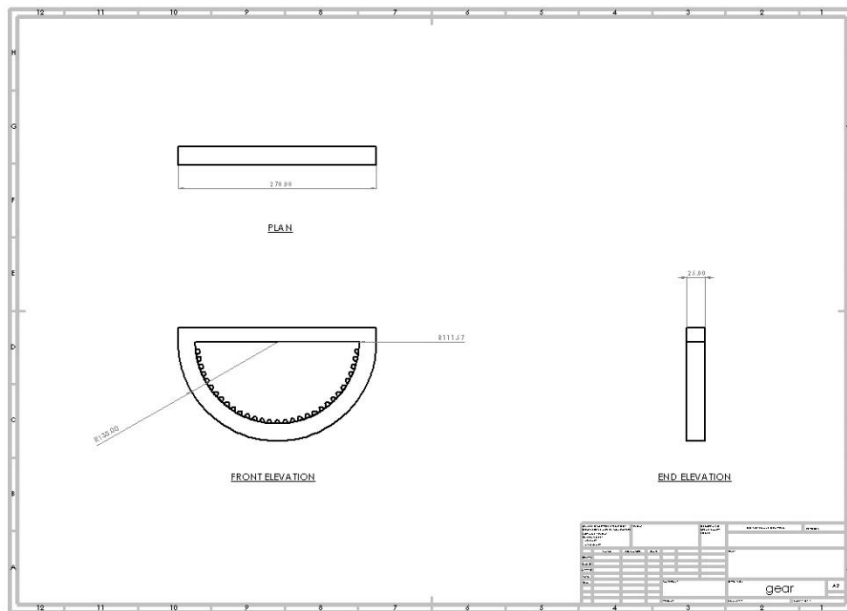
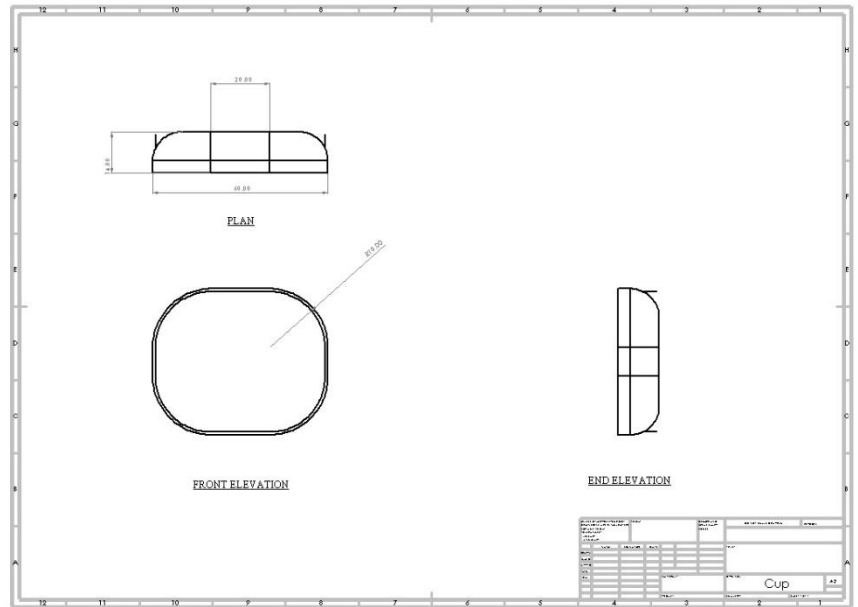
- Concept F was the best-fit design following a rigorous evaluation process based on the key performance criteria such as adjustability, stability, durability, and usability. Its bevel gear-based height adjustment system allows smooth and precise adjustment with reduced effort, offering an ergonomic and user-friendly solution. The toothed bar angle adjustment mechanism also has a locking mechanism that prevents accidental movement and stays stable when in use. These design elements give Concept F an extremely practical and reliable workstation aptly suited for various user usage, including office work, study, and creative work.

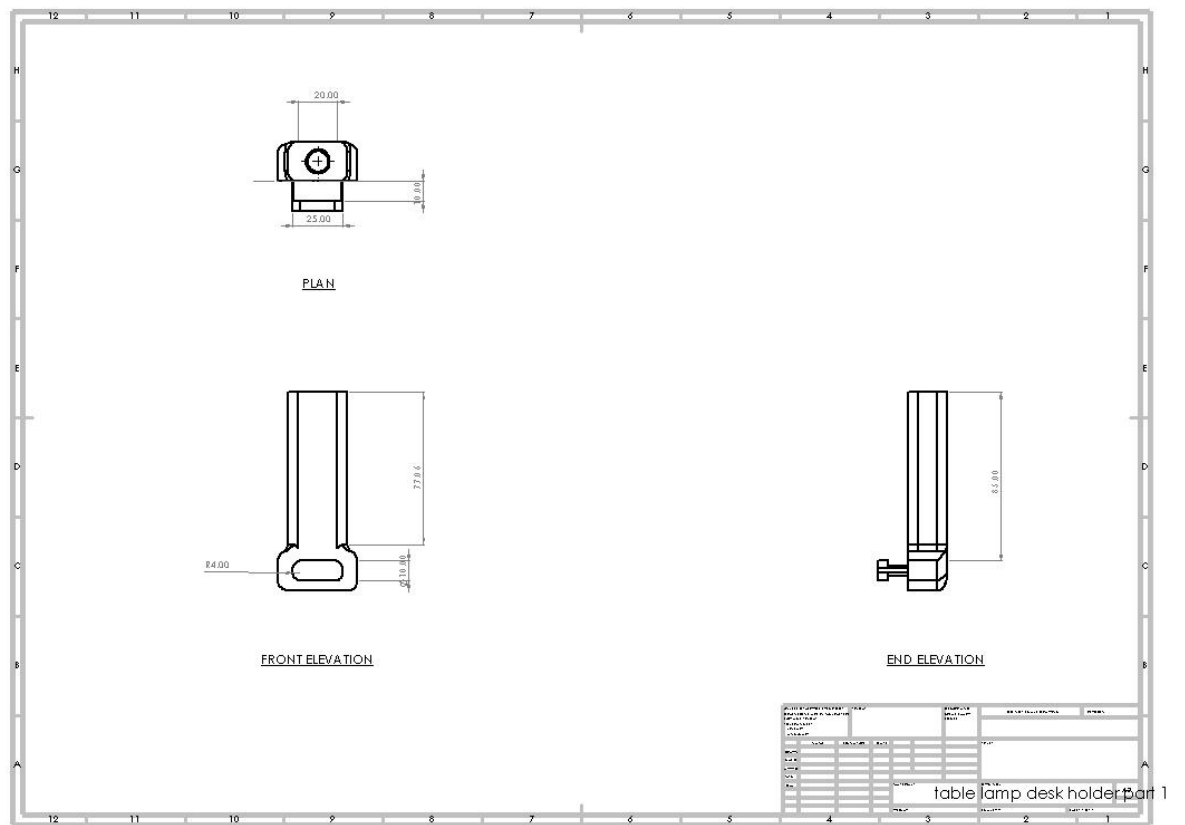
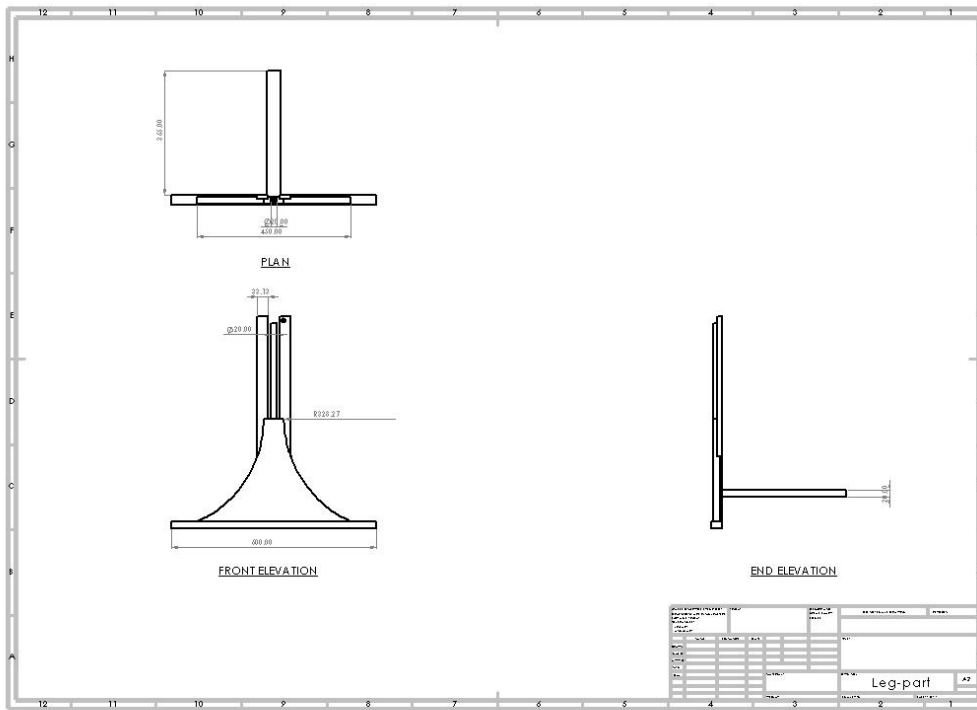
❖ 3D Model

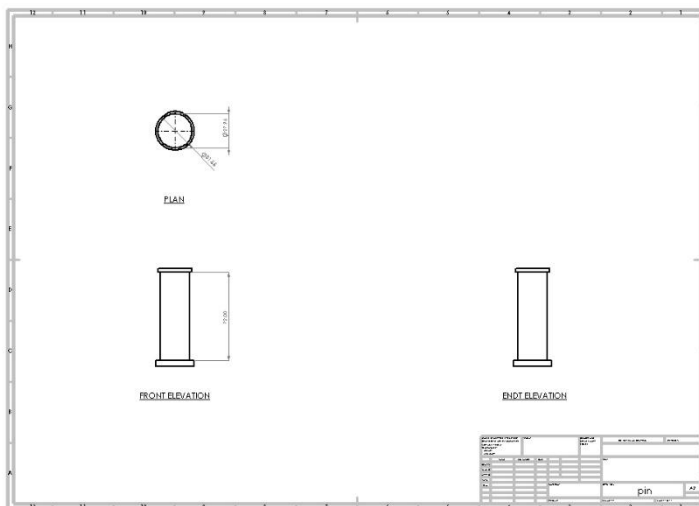
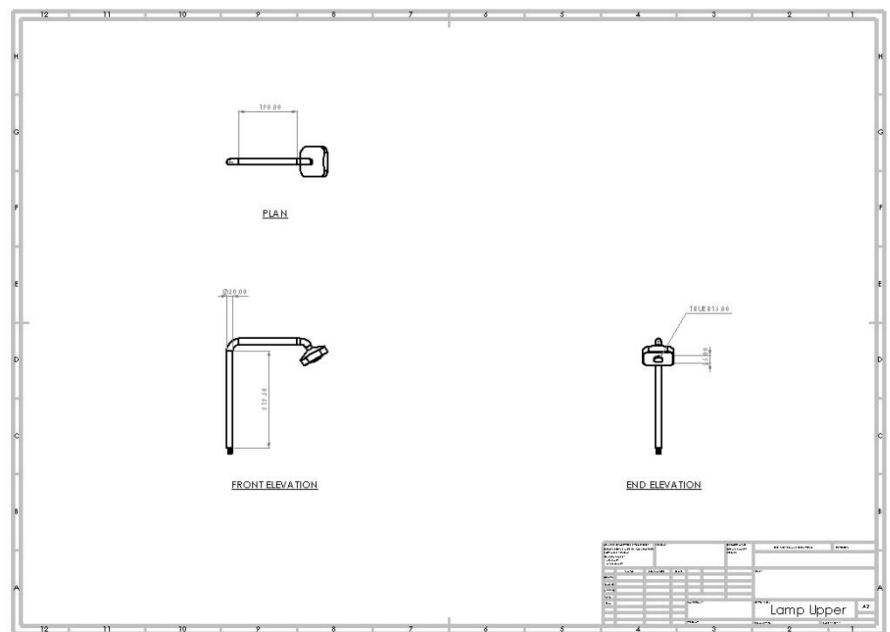
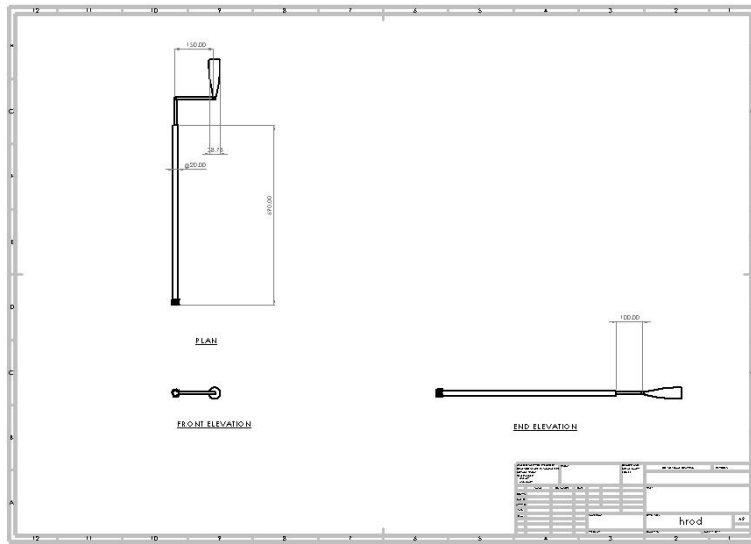


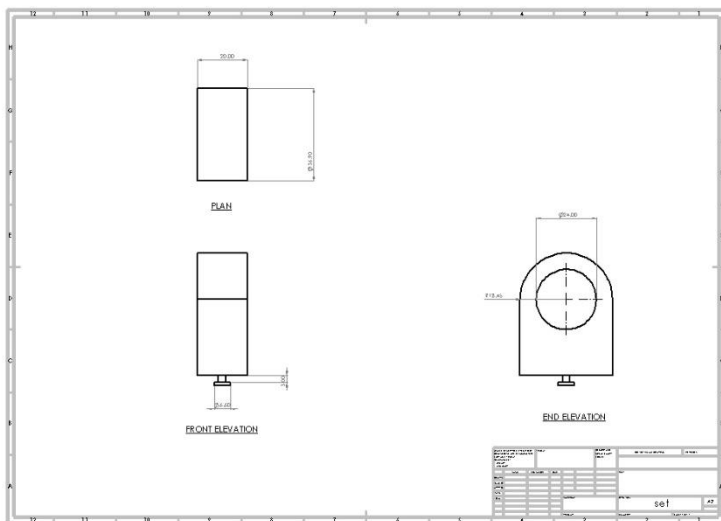
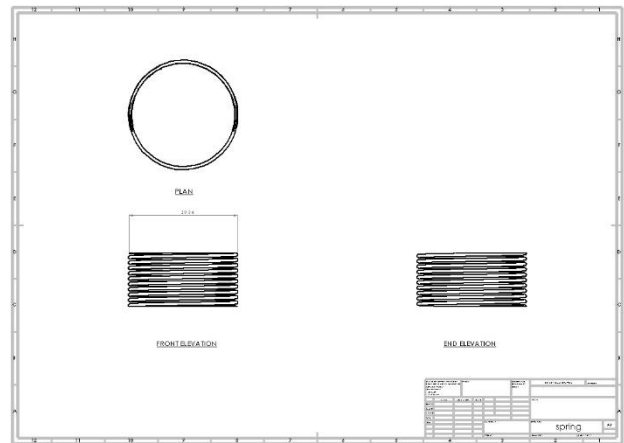
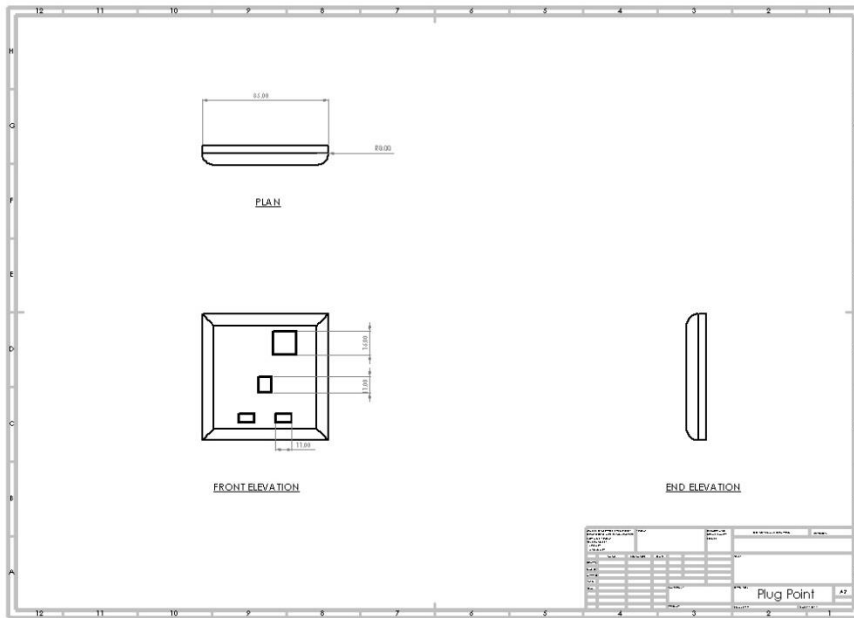
❖ 2D Parts

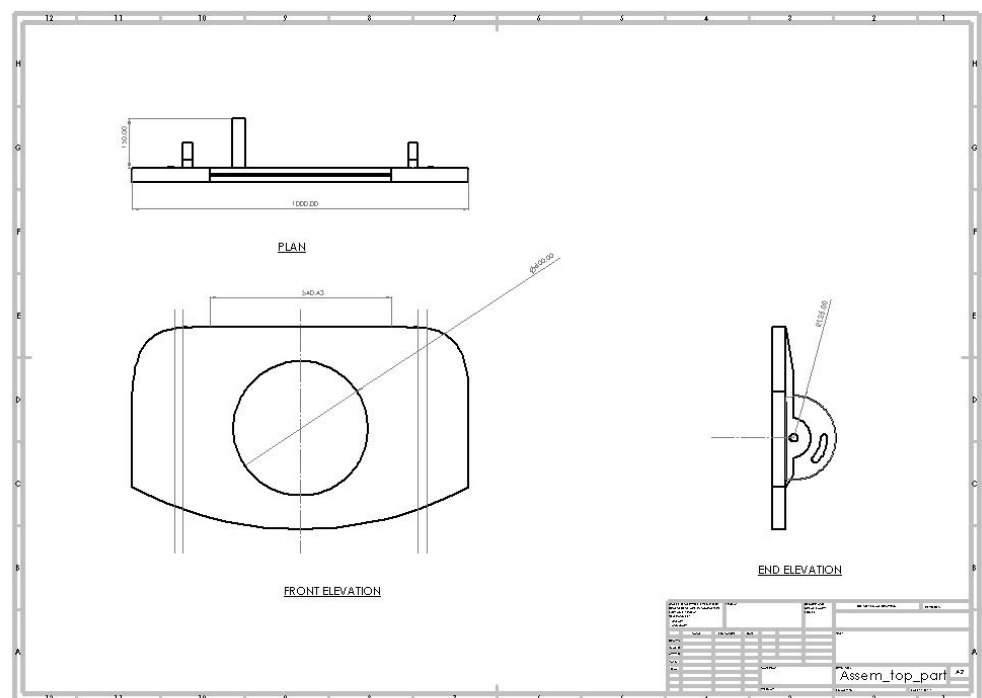
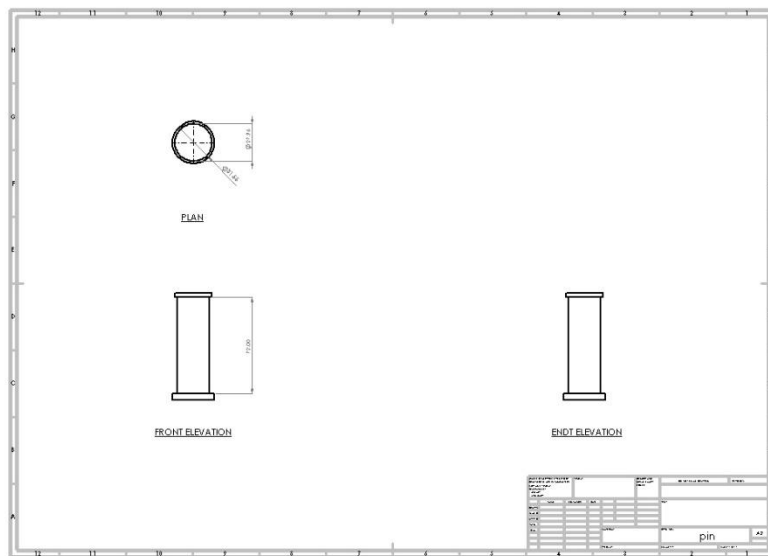












● CONCLUSION

- The project focused on designing a height and angle adjustable desk to improve ergonomics, comfort, and productivity for users. After identifying the market opportunity, analyzing customer needs, and specifying product requirements, several design concepts were generated and evaluated.
- Through a structured selection process, Concept F was chosen as the best design due to its bevel gear-based height adjustment system and toothed bar angle adjustment mechanism, offering smooth control, stability, and durability. The final design was optimized to enhance usability, ensuring it met the key performance criteria.
- The outcome of the work includes a refined design, 3D models, and 2D technical drawings, demonstrating the feasibility of a practical, ergonomic workstation suitable for various user needs, including office work, study, and creative activities.



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